

SANDEEP B

+91-8050868752 ◇ Bengaluru, India

Email: bsandeep527@gmail.com ◇ [Webpage](#) ◇ [Linkedin](#)

EDUCATION

Master of Science by Research (MSR), [Indian Institute of Technology Bombay](#) Dec'2021 - Jan'2024
Mechanical Engineering - Design Specialization, GPA: 7.5/10
Mumbai, India

Master of Technology (M.Tech), [Manipal Academy of Higher Education](#) 2017 - 2018
[Manipal Institute of Technology, Manipal, India](#)

Mechatronics Engineering - Industrial Automation and Robotics Specialization, GPA: 8.11/10
[Indian Institute of Technology Kanpur, India](#) 2018 - 2019
Exchange Student - Electrical Engineering

Bachelor of Engineering (B.E), [Visvesvaraya Technological University](#) 2012 - 2016
[St Joseph Engineering College, Mangalore, India](#)
Mechanical Engineering, Percentage: 63.57

AREA OF INTERESTS

Research & Development, Finite Element Methods and Analysis, Non-Destructive Testing, New Product Development, Ultrasonic Guided Wave, Stress Analysis, Structural Health Monitoring, Ansys, CAD.

EXPERIENCE

Project Engineer July 2019 - June 2021
Indian Institute of Technology Kanpur *Kanpur, India*

Remote Metrology of Hexagonal Wrapper Tube through Glass Medium. ([view here](#))

- The project aims to develop a damage inspection profilometry system for a hexagonal subassembly component of a nuclear fast breeder reactor core using a structured light source/and an imaging sensor that operates through a radiation shield glass medium 80-120 cm thick, placed between the component and the source and imaging system.
- The project involves developing, standardising, and implementing the methodology for remote metrology of a hexagonal wrapper (of 500mm length) positioned at about 2.0-2.5m distance from the laser/camera system and through the shielding glass window. Expected precision 50 microns.

Sheet Counting. Client: TATA Tinplate. ([view here](#))

- The machine visually counts stacked in the range of 200-micron thickness sheets with high accuracy (<0.1%). In our case, they are thin tin-coated mild steel sheets with optically challenging specular characteristics. The sheets are further not expected to be perfectly aligned. Manually counting such sheets is dangerous and time-consuming.
- A naive way of counting stacked substrate would be to divide the overall thickness/weight by the nominal thickness/weight of one sheet in the stack. However, this method fails to achieve the required accuracy because individual sheets deviate in their thickness/weight from the nominal values because of manufacturing variations, which could lead to counting errors more significant than 1%. The machine can count a 50 cm stack in under 3 minutes. The system includes image acquisition, optical distortion correction, customised stitching, and occluded sheet compensation. The counting result is displayed on an included numeric display. (Complete Hardware Development).

Bottle Inspection. ([view here](#))

- The system must inspect the soft drink PET bottles and automatically discard the faulty bottle from the conveyor. The bottle sizes vary from 250 ml to 1.5 litres. The faults of the bottle are under and overfilled liquid, damaged closure, and torn labels.

- A single high-speed camera captures images on the conveyor; 400-600 bottles per minute is the conveyor's travel speed. The captured images are processed; the default bottles are kicked away from the conveyor using a pneumatic system.

Significant Responsibilities: Product Development, Mechanical Design, 3D model development, and prototyping.

Teaching Assistant

Indian Institute of Technology Bombay

March 2023 - June 2023

Mumbai, India

- Head of the Teaching Assistant group.
- Holding full accountability for delivering tutorial sessions and administering quizzes and assignments for 120 students.

DISSERTATION WORK

Damage Detection in Insulated Pipes Using Ultrasonic Guided Waves - Master of Science by Research - under the guidance of [Professor Nitesh P. Yelve](#), IIT Bombay. ([view here](#))

- Pipes structures are widely spread and used in the industrial sectors, mainly oil, gas, chemical, etc. These pipes travel for hundreds of kilometres under the earth's surface; they become inaccessible and difficult to inspect at every location. Also, it isn't easy to inspect the outer surface of the pipe as they are insulated from the outer surface.
- Structural Health Monitoring (SHM) using the Ultrasonic Guided Waves (UGW) method is adapted to monitor the structure. UGW testing is used to locate and localise the presence of damage.

Techniques for Miniature Actuation and Imaging (Engine Inspection) - Master of Technology - under the guidance of [Professor K. S. Venkatesh](#), IIT Kanpur and Vijay Kumar Pandey (PhD). ([view here](#))

- Diesel oil engine inspection should be achieved without stripping down the system to save inspection time. The fuel injector has two different configurations: the injector is parallel to the bore axis or inclined at a specified angle to the bore cylinder axis.
- A borescope camera inspects the 360° angular extent of the bore aligned axially with the bore but fitted with a mirror of 45° to obtain a view of the lateral surface of the bore. The images/video captured from the camera are mosaiced and displayed on the screen as an opened cylinder.

Investigation on the Performance and Emissions Characteristics of IC engines using different blends of Waste Cooking Oil Methyl Ester-Ethanol-Diesel Oil. ([view here](#))

- The experiment was conducted by keeping a constant compression ratio and varying speeds; significantly improved brake thermal efficiency and brake-specific fuel consumption were observed when compared with diesel oil.
- NO_x, CO, and CO₂ emissions for ethanol blends are less than the diesel oil at higher loads.

ACADEMIC PROJECTS

Machine Learning Projects:

- Coded and tested gradient descent to train and validate L1 and L2 penalised linear regression from scratch.
- Performed exploratory data analysis and eliminated correlated variables and imputed missing variables. I also trained and validated hyperparameters of elastic net logistic regression, neural networks, support vector machines, and random forests for a classification problem. I also checked feature importance and eliminated variables to improve model performance.
- Modified the pre-existing CNN training code to classify bees versus ants to run on Google CoLab to test learning rate and momentum and added weight decay for regularisation. I also pre-processed data for clustering using various transforms and trained and visualised the effect of k in k-means clustering. I also trained in principal component analysis and selected the number of dimensions based on the variance explained. I also trained kernel principal component analysis and selected the number of dimensions based on the variance explained.

Gesture Control of EMU two link Manipulator (Mini Project)

- Controlling the EMU robot with a hand gesture using the Inertial measurement unit sensors.

Smart switch (Mini Project)

- IoT-enabled presence-activated smart lighting solution for use in the Mechatronics department of Motion Tracking (Mini Project)

SKILLS

Software and Tools	MATLAB, Catia V5, AutoCAD, Ansys, Abaqus, Solid Edge, Arduino IDE, LaTeX, Collab, Ultimaker Cura.
Language	Python, PLC.
Operating System	Linux, Microsoft Windows, and MacOS.

MASTER'S COURSES

Introduction to Machine Learning, Finite Element Methods, Mathematics for Simulation and Technical Modelling, Sensors and Actuators for Industrial Automation, Introduction to Industrial Robots, Analog and Digital Electronics, Signal Processing and Applications, Fluid Power Systems and Factory Automation, Embedded Systems, Robot Dynamics and Analysis, Machine Vision and Image Processing, Artificial Intelligence and Expert Systems in Auto, Renewable Energy Technology.

Labs: PLC and Sensors, Drives and Controls, Advanced Robotics, Hydraulics, and Pneumatics.

CERTIFICATIONS

- [Diploma in Product Design \(Catia V5, AutoCAD, and GD & T\) by CADD Centre Training Services.](#)
- Modern Robotics, Foundations of Robot Motion, an online non-credit course offered by Northwestern University through Coursera.
- Convolutional Neural Networks is an online non-credit course authorised by IBM and offered through Coursera.
- Neural Networks and Deep Learning is an online non-credit course authorised by deeplearning.ai and offered through Coursera.
- Introduction to Machine Learning, an online course by Duke University and offered through Coursera.
- Programming for Everybody (Getting Started with Python) is an online course offered by the University of Michigan through Coursera.

TRAINING

- Beckhoff Automation Private Limited organised TwinCAT Software.
- Graphical System Design and virtual Labs using LabVIEW organised by the National Institute of Technology Suratkal.
- Computational Fluid Dynamics by HT India Labs.

PUBLICATION

V. Kaliveer, P. R. Sequeira, R. V. D, S. A. Dsilva, B. Sandeep, and S. D. Rolvin, "Investigation on the Performance and Emissions Characteristics of CI Engine Using Different Blends of Waste Cooking Oil Methyl Ester-Ethanol-Diesel Oil," vol. 6, pp. 28–32, 2016.

AWARD

- Best Student Award by Indian Society for Technical Education (ISTE) for Karnataka Section, 2016.

LANGUAGES

English (Full professional proficiency)	Kannada (Native proficiency)
Hindi (Bilingual proficiency)	Telugu (Limited working proficiency)
Tulu (Limited Working proficiency)	German (Elementary proficiency (A1))

REFEREES

Professor K S Venkatesh, Electrical Engineering Department, Indian Institute of Technology Kanpur, Kanpur, India. venkats@iitk.ac.in

Professor Nitesh P. Yelve, Mechanical Engineering Department, Indian Institute of Technology Bombay, Mumbai, India. nitesh.yelve@iitb.ac.in

Professor Binu K.G, Mechanical Engineering Department, St Joseph Engineering College, Mangaluru. binuk@sjec.ac.in